

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – Industry based Problem-Solving Question

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4-Industry Based Problem	Assessment:	Assessment Tool 2– Industry Based Problem
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and apply statistical inference techniques, including point estimation, confidence intervals, and hypothesis testing, to interpret sample data and support effective decision-making. (BL4)		
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Section / Batch:	CSE AI 2024	Date:	19-08-2026

Code of Conduct

I Y SRAVAN KUMAR (192472289) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: SRAVAN

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

CO4-AT2 Industry Problem Solving

1. Digital Marketing – Hypothesis Testing Using Confidence Interval

Question

Before introducing a new advertisement, the conversion rate was 8%. After the campaign, 90 out of 1,000 visitors converted. Using a 95% confidence interval, determine whether the observed conversion rate provides evidence of improvement.

Given

Number of visitors:

$$n = 1000$$

Number of conversions:

$$x = 90$$

Previous conversion rate:

$$p_0 = 0.08$$

Observed conversion rate:

$$\hat{p} = x / n$$

$$\hat{p} = 90 / 1000$$

$$\hat{p} = 0.09 = 9\%$$

Confidence level = 95%

For 95% confidence level:

$$z = 1.96$$

Step 1: Calculate Standard Error

$$SE = \sqrt{[\hat{p}(1 - \hat{p}) / n]}$$

$$SE = \sqrt{[0.09(1 - 0.09) / 1000]}$$

$$SE = \sqrt{[0.09(0.91) / 1000]}$$

$$SE = \sqrt{[0.0819 / 1000]}$$

$$SE \approx 0.00905$$

Step 2: Calculate Confidence Interval

$$CI = \hat{p} \pm z(SE)$$

$$CI = 0.09 \pm 1.96(0.00905)$$

$$CI = 0.09 \pm 0.01774$$

$$\text{Lower limit} = 0.09 - 0.01774$$

$$\text{Lower limit} = 0.07226$$

$$\text{Upper limit} = 0.09 + 0.01774$$

$$\text{Upper limit} = 0.10774$$

Therefore:

$$\text{CI} = (0.07226, 0.10774)$$

$$\text{CI} \approx (7.23\%, 10.77\%)$$

Step 3: Interpretation

Previous conversion rate = 8%

$$95\% \text{ CI} = (7.23\%, 10.77\%)$$

Since **8% lies inside the confidence interval**, the previous conversion rate is consistent with the observed data.

Conclusion

The observed conversion rate increased from **8% to 9%**, but the 95% confidence interval does **not provide sufficient evidence of a statistically significant improvement**.

2. Education Technology – Hypothesis Testing Using Confidence Interval

Question

An EdTech company claims that its adaptive-learning platform improves the average test score to at least 75. A sample of 64 students obtains an average score of 78 with standard deviation 10.

Use a 95% confidence interval to assess the claim.

Given

Sample size:

$$n = 64$$

Sample mean:

$$\bar{x} = 78$$

Sample standard deviation:

$$s = 10$$

Claimed mean:

$$\mu_0 = 75$$

Confidence level = 95%

$$z = 1.96$$

Step 1: Calculate Standard Error

$$SE = s / \sqrt{n}$$

$$SE = 10 / \sqrt{64}$$

$$SE = 10 / 8$$

$$SE = 1.25$$

Step 2: Calculate Confidence Interval

$$CI = \bar{x} \pm z(SE)$$

$$CI = 78 \pm 1.96(1.25)$$

$$CI = 78 \pm 2.45$$

$$\text{Lower limit} = 78 - 2.45$$

$$\text{Lower limit} = 75.55$$

$$\text{Upper limit} = 78 + 2.45$$

$$\text{Upper limit} = 80.45$$

Therefore:

$$CI = (75.55, 80.45)$$

Step 3: Interpretation

Claimed minimum average score = 75

$$95\% CI = (75.55, 80.45)$$

The **entire confidence interval is above 75.**

Conclusion

The 95% confidence interval supports the EdTech company's claim that the true average test score is **at least 75.**

3. Automobile Industry – Hypothesis Testing Using p-value

Question

A car manufacturer claims that its new model provides an average mileage of 20 km/litre. A sample of 50 cars gives a mean mileage of 19.2 km/litre with standard deviation 2.5 km/litre. Perform a hypothesis test at the 5% significance level using the p-value approach.

Given

Sample size:

$$n = 50$$

Sample mean:

$$\bar{x} = 19.2 \text{ km/litre}$$

Sample standard deviation:

$$s = 2.5 \text{ km/litre}$$

Claimed mean:

$$\mu_0 = 20 \text{ km/litre}$$

Significance level:

$$\alpha = 0.05$$

Step 1: State the Hypotheses

$$H_0: \mu = 20$$

$$H_1: \mu \neq 20$$

This is a **two-tailed test**.

Step 2: Calculate Standard Error

$$SE = s / \sqrt{n}$$

$$SE = 2.5 / \sqrt{50}$$

$$SE = 2.5 / 7.071$$

$$SE \approx 0.3536$$

Step 3: Calculate Test Statistic

$$z = (\bar{x} - \mu_0) / SE$$

$$z = (19.2 - 20) / 0.3536$$

$$z = -0.8 / 0.3536$$

$$z \approx -2.26$$

Step 4: Calculate p-value

For $z = -2.26$, the two-tailed p-value is approximately:

$$p\text{-value} \approx 0.024$$

Step 5: Compare p-value with α

$$p\text{-value} = 0.024$$

$$\alpha = 0.05$$

$$0.024 < 0.05$$

Therefore:

Decision = Reject H_0

Conclusion

There is statistically significant evidence that the actual average mileage is **different from 20 km/litre**.

Since the sample mean is 19.2 km/litre, the data suggests that the actual mileage is **lower than the claimed 20 km/litre**.

Therefore, the manufacturer's claim is **not supported at the 5% significance level**.

4. Cloud Computing – Hypothesis Testing Using p-value

Question

A cloud provider claims that its average server response time is 100 ms. A sample of 81 requests has a mean of 104 ms and standard deviation of 12 ms. Test the claim at $\alpha = 0.05$ using the p-value method.

Given

Sample size:

$$n = 81$$

Sample mean:

$$\bar{x} = 104 \text{ ms}$$

Sample standard deviation:

$$s = 12 \text{ ms}$$

Claimed mean:

$$\mu_0 = 100 \text{ ms}$$

Significance level:

$$\alpha = 0.05$$

Step 1: State the Hypotheses

$$H_0: \mu = 100$$

$$H_1: \mu \neq 100$$

This is a **two-tailed test**.

Step 2: Calculate Standard Error

$$SE = s / \sqrt{n}$$

$$SE = 12 / \sqrt{81}$$

$$SE = 12 / 9$$

$$SE = 1.3333$$

Step 3: Calculate Test Statistic

$$z = (\bar{x} - \mu_0) / SE$$

$$z = (104 - 100) / 1.3333$$

$$z = 4 / 1.3333$$

$$z \approx 3.00$$

Step 4: Calculate p-value

For $z = 3.00$, the two-tailed p-value is approximately:

$$\text{p-value} \approx 0.0027$$

Step 5: Compare p-value with α

$$\text{p-value} = 0.0027$$

$$\alpha = 0.05$$

$$0.0027 < 0.05$$

Therefore:

Decision = Reject H_0

Conclusion

There is statistically significant evidence that the average server response time is **different from 100 ms**.

Since the observed average is 104 ms, the data suggests that the actual response time is **higher than 100 ms**.

Therefore, the cloud provider's claim is **not supported at the 5% significance level**.

5. E-commerce – Hypothesis Testing Using p-value

Question

An online company claims that its product return rate is 10%. In a sample of 1,000 orders, 130 products were returned. Test whether the actual return rate is significantly higher than 10% using the p-value approach at the 5% significance level.

Given

Sample size:

$$n = 1000$$

Number of returned products:

$$x = 130$$

Claimed return rate:

$$p_0 = 0.10$$

Significance level:

$$\alpha = 0.05$$

Step 1: Calculate Observed Return Rate

$$\hat{p} = x / n$$

$$\hat{p} = 130 / 1000$$

$$\hat{p} = 0.13$$

$$\hat{p} = 13\%$$

Step 2: State the Hypotheses

$$H_0: p = 0.10$$

$$H_1: p > 0.10$$

This is a **right-tailed test** because we want to determine whether the return rate is higher than 10%.

Step 3: Calculate Standard Error

$$SE = \sqrt{[p_0(1 - p_0) / n]}$$

$$SE = \sqrt{[0.10(1 - 0.10) / 1000]}$$

$$SE = \sqrt{[0.10(0.90) / 1000]}$$

$$SE = \sqrt{[0.09 / 1000]}$$

$$SE \approx 0.00949$$

Step 4: Calculate Test Statistic

$$z = (\hat{p} - p_0) / SE$$

$$z = (0.13 - 0.10) / 0.00949$$

$$z = 0.03 / 0.00949$$

$$z \approx 3.16$$

Step 5: Calculate p-value

For $z = 3.16$, the right-tailed p-value is approximately:

$$\text{p-value} \approx 0.0008$$

Step 6: Compare p-value with α

$$\text{p-value} = 0.0008$$

$$\alpha = 0.05$$

$$0.0008 < 0.05$$

Therefore:

Decision = Reject H_0

Conclusion

There is strong statistical evidence that the actual product return rate is **significantly higher than 10%**.

The observed return rate is 13%, compared with the claimed rate of 10%.

Therefore, the company should conclude that its actual product return rate is **significantly higher than 10% at the 5% significance level**.